

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Substitution into: $PE = mgh$ i.e. $750 = 3 \times 10 \times h$ (1) Rearrangement: $h = \frac{750}{3 \times 10}$ (1) $h = 25$ [m] (1)	1	1		3	2	
		(ii)		Substitution into: $KE = \frac{1}{2} mv^2$ i.e. $600 = \frac{1}{2} \times 3 \times v^2$ (1) Rearrangement: $v^2 = \frac{2 \times 600}{3}$ (1) $v = 20$ [m/s] (1)	1	1		3	2	
		(iii)		$[750 - 600] = 150$ [J]		1		1	1	
		(iv)		Substitution into $WD = F \times d$ i.e. $150 \text{ ecf} = F \times 25 \text{ ecf}$ (1) Rearrangement: $F = \frac{150}{25}$ (1) $F = 6$ [N] (1)	1	1		3	2	
	(b)	(i)		Forces named: weight and {air resistance / drag} (1) NOT gravity Air resistance increases as speed increases (1) When the forces are {balanced / equal and opposite} OR resultant force becomes zero [terminal speed is reached] (1)	3			3		
		(ii)		<u>Ball pulls</u> [equally] <u>upwards on the Earth</u> (1) <u>Air pushes</u> [equally] <u>upwards on the ball</u> (1)		2		2		
				Question 4 total	6	9	0	15	7	0